DATA ITEM DESCRIPTION

Title: MACHINERY AND EQUIPMENT PRE-INSTALLATION AIRBORNE SOUND MEASUREMENTS TEST REPORT

Number: DI-HFAC-80272A
AMSC Number: N9535
DTIC Applicable: No
Preparing Activity: SH
Applicable Forms:

Approval Date: 20150413
Limitation: N/A
GIDEP Applicable: No
Project Number: HFAC-2015-007

Use/relationship: The Machinery and Equipment Pre-Installation Airborne Sound Measurements Test Report provides a description of the airborne sound tests and results in order to demonstrate compliance with the requirements of MIL-STD-1474.

a. This Data Item Description (DID) contains the format, content, and preparation instructions for that data generated as specified in MIL-STD-1474 and ANSI/ASA S12.67, *Pre-Installation Airborne Sound Measurements and Acceptance Criteria of Shipboard Equipment*.

b. This DID shall be applied for each machinery and equipment item with an airborne noise requirement, excluding fans, fan-coil units, fan-coil assemblies, air handling units, and other air moving devices when MIL-STD-1474 and ANSI/ASA S12.67 is invoked in the contract specifications or statement of work.

c. The data reported by this DID may be used to support the analysis and design development described in DI-HFAC-81278, *Airborne Noise Analysis and Control Design History Booklet*.

d. This DID supersedes DI-HFAC-80272.

Requirements:

1. Reference documents. The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices, and revisions shall be as specified in the contract.

2. Format. The Machinery and Equipment Pre-Installation Airborne Sound Measurements Test Report shall be in the Contractor's format.

3. Content. The Machinery and Equipment Pre-Installation Airborne Sound Measurements Test Report shall contain the following sections.

3.1. Test and equipment information. This section shall contain:

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a. Make, model, and serial number of the unit under test, including identification of major components, driver, and driven unit.
b. Ship name(s) and hull number(s) for which the unit is being acquired.
c. Compartment or on-deck location where the unit will be installed. If identical units by make and model number will be installed in any one compartment, identify the specific locations in that compartment where the unit will be installed.
d. Overall unit dimensions and orientation of machinery or equipment item undergoing testing.
e. Evaluating activity and test site, including name, telephone number, and e-mail addresses of the individuals who performed and supervised the tests.
f. Test date.
g. Specified acceptance criteria, including reference units.
h. Driver unit horsepower.
i. Rated electrical characteristics, including volts, amps, and phase.
j. Rated fluid media parameters (e.g., flow rates, temperatures, pressures).
k. Equipment rated operational speeds (e.g., driver and driven unit revolutions per minute (rpm)).
l. Expected equipment primary source excitation frequencies (e.g., rotational frequencies, gear tooth mesh frequencies, motor slot frequency, cylinder firing frequencies, vane frequencies, blade frequencies, piston frequencies, turbocharger frequencies).
m. Resilient mount manufacturer, make, model number, date of manufacture, load range, and natural frequency of resilient mounts used to support the device under test.
n. Number of resilient mounts used. A photo shall be included showing each installed resilient mount.
o. If the use of non-Navy standard resilient mounts has been approved, include the characteristics including load rating in pounds and resonant frequency of the mounts.
p. Operational load (e.g., full, half) of the machinery or equipment undergoing testing.
q. Operating speeds (rpm) used during test.
r. Throttling conditions, if any, and a description of how throttling conditions are set an verified.
s. Operating suction and discharge fluid media parameters during test (e.g., flow rates, pressures, temperatures).
t. Electrical inputs, outputs, or both used during test.
u. Identification of make and model of measurement and analysis instrumentation used, including calibration instrumentation. Certificates of laboratory calibration shall be reported.
v. Details of instrument settings used during the testing, e.g., type of sampling window, coupling, sampling frequency, sampling time, number of averages, averaging type, and frequency resolution of the data, if appropriate.

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w. Name and other information identifying any individuals representing the procuring activity who witnessed the test, including e-mail and telephone number.

3.2. Drawings, photographs, and calculations. This section shall contain:

a. Identification and arrangement of major components and unit under test.
b. Resilient mount location and orientation, including any sway braces; and overall wet weight and center of gravity of the item undergoing testing with respect to a defined reference position.
c. Calculated load on each resilient mount compared to specified load range of each mount; and calculated natural frequency of each mount when experiencing the calculated load.
d. Drawing and dimensional description of the foundation and/or mounting structure used to support the machinery or equipment item undergoing testing.
e. Arrangement and description of flexible hose assemblies and flexible duct connections showing configuration, diameter and lengths of flexible sections, and details of how the weights of these were supported. Photos shall be included.
f. Arrangement of piping and fittings used to facilitate testing and details of how the weights of these were supported.
g. Arrangement of electrical power leads coupled with arrangements of cables used with instrumentation.
h. Location and orientation of each microphone relative to the envelope of the item under test.
i. Locations and sizes of equipment air inlet and discharge openings.
j. Location of acoustical center of sound source.
k. Acoustical environment (indoor). When testing indoors, the following shall be included:
    (1) Location and orientation of test unit relative to floor, walls, ceiling, and contents of test room.
    (2) Test location (e.g., anechoic, reverberant); dimensions of the test room or chamber; description of the physical treatment on walls, ceiling, and floor.
    (3) Description of any exposed insulating materials installed on the walls, floor, or ceiling including manufacturer, model number, thickness, density, and facing.
l. Acoustical environment (outdoor). When testing outdoors, the following shall be included:
    (1) Location and orientation of test unit relative to buildings, structures or other reflecting surfaces or objects.
    (2) Description of the ground surface, and wind direction and speed.

3.3. Measured data. This section shall contain:

a. Measured data in reference units that are identical to the specified reference units.

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b. Any measurements that are included in graphical form shall show ordinate (magnitude) values with 10 dB between major divisions.
c. Measured data compared, by measurement location and orientation, with accompanying information at that same location including background and ambient data and assigned limits.
d. For airborne noise levels that exceed the specified limits, excesses shall be clearly indicated in any tabulated and plotted data.
e. If excesses to the specified octave band airborne levels are included in graphical form, plots of narrowband data measured at the offending microphone positions, with frequency resolution not to exceed 1 Hz, shall be included.
f. The report shall identify the mechanisms that exist within the machinery or equipment item that are specifically responsible for the measured excesses (e.g., rotational frequency, blade, vane or piston frequency, firing frequency, gear mesh frequency, coincidence of a forcing frequency with a natural frequency of the mounting system, resonance in the subbase).

End of DI-HFAC-80272A

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